Raghavendra Kulkarni

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Education

Northeastern University — Boston, MA
Master of Science in Computer Science (GPA: 3.86) — *Aug. 2024 – Dec. 2026*

Visvesvaraya Technological University — Bengaluru, India
Bachelor's in Information Science and Engineering — *Aug. 2018 – July 2022*

Experience

Software Engineer Intern — Jan. 2026 – Aug. 2026
Vicor Corporation — *Andover, MA*

- Engineered **Java/Spring Boot REST APIs** to integrate SAP ERP and Oracle PLM data into a distributed core engineering platform, implementing payload caching and batch processing to seamlessly handle **10,000+** daily transactions.
- Optimized an engineer-facing PCB workflow by rewriting **React/JavaScript** components with asynchronous data fetching and lazy loading, cutting workflow latency from **80s to 2s** and boosting productivity for **200+** internal engineers.
- Designed automated **CI/CD** pipelines using **GitHub Actions and Docker** for containerized applications, cutting cross-environment deployment times by **45%** while ensuring reliable feature delivery at scale.
- Refactored tightly coupled service and DAO layers using modern structural design patterns; engineered robust **JUnit 5/Mockito** test suites to achieve **95%+** coverage, reducing production regression defects by **50%**.

AI Software Engineer — Sep. 2025 – Dec. 2025; Sep. 2026 – Present
DASH AI Hub, Northeastern University — *Boston, MA*

- Engineered a full-stack AI evaluation platform (Ava) using **React, Next.js, TypeScript, and Flask**, translating complex requirements into scalable REST APIs that slashed workflow processing time from **8+ hours to under 30 minutes** for **300+** active users.
- Architected production-grade RAG and agentic orchestration pipelines utilizing **LangGraph, LlamaIndex, and Qdrant**, optimizing vector retrieval and semantic search frameworks to ensure high-accuracy, hallucination-resistant LLM outputs.
- Built a scalable multi-agent voice coaching platform (Lyra) using **TypeScript, FastAPI, and PostgreSQL**, implementing stateful session management, strict API contracts, and complex LLM-to-LLM feedback routing for persistent user interactions.
- Orchestrated **asynchronous distributed backend services** via **Celery, RabbitMQ, and Redis** to manage intensive GPU inference workloads, leveraging robust task queuing and retry logic to maintain highly responsive frontend APIs.

Software Engineer — Oct. 2022 – Aug. 2024
Cognizant Technology Solutions — *Bengaluru, India*

- Engineered core **React** components and **Java/Spring Boot** REST APIs for a multi-cloud infrastructure platform; optimized state management and payload caching to cut workflow latency by 60% across **AWS, Azure, and SAP HANA Cloud**.
- Automated container deployments for 15+ microservices using **Terraform, Kubernetes, and Helm**, configuring dynamic scaling policies that cut cross-environment spin-up times from days to under 45 minutes.
- Architected asynchronous **Python** backend integrations and robust automation workflows for enterprise routing, eliminating **40+** hours of manual setup per environment to securely onboard **500+** enterprise tenants.
- Designed blue-green CI/CD pipelines via **Jenkins** and **GitHub Actions** for a large-scale distributed system, integrating **Grafana** and **Splunk** telemetry to accelerate release cadence 3x and reduce p99 incident response times.

Software Engineer Intern — *Apr. 2022 – Sep. 2022*
Cognizant Technology Solutions — *Bengaluru, India*

- Migrated full-stack modules from **.NET to Java and Angular** and optimized **SQL** queries in a high-traffic distributed system, cutting page latency by **40%** and eliminating recurring timeouts for **300K+** users.

Projects

Learnbot | *React, Flask, Pinecone, Claude, RAG*

- Engineered a **React/Flask** RAG based AI tutor using **Claude** and **Pinecone** vector search to process unstructured Canvas data.
- Deployed the application for **50+ students**, implementing strict prompt engineering and semantic guardrails for accuracy.

PlanVoyage | *React, Node.js, PostgreSQL, Docker, AWS*

- Architected a full-stack **React/Node.js** trip-planning application with secure **JWT** authentication and **PostgreSQL**.
- Containerized microservices via **Docker** and engineered **GitHub Actions CI/CD** pipelines for automated **AWS** deployments.

Technical Skills

Languages: Java, Python, JavaScript, TypeScript, SQL
Frameworks: Spring Boot, React, Next.js, Angular, Flask, FastAPI, Node.js, JPA/Hibernate, JUnit, Mockito
AI/ML & Databases: NLP, LangGraph, LlamaIndex, LangChain, Qdrant, Pinecone, PostgreSQL, MongoDB, Oracle, Redis
Cloud & Infrastructure: AWS, Azure, Docker, Kubernetes, Terraform, RabbitMQ, Jenkins, GitHub Actions, Linux, Git
Core Competencies: Data Structures & Algorithms, System Design, Microservices, Distributed Systems, Version Control, SDLC, Agile
AI Tools: Claude Code, Codex, Github Copilot, Open-source LLMs, MCP